$$\begin{aligned}
\mu_\theta &= \frac{-0.2 - (0.01/0.6)(-0.9)}{\sqrt{0.99}} \\
&\approx -0.18593 \\
x_{t-1} &= -0.18593 + 0.1 \cdot 0.3 \approx -0.15593
\end{aligned}$$